IE 412 Homework 2

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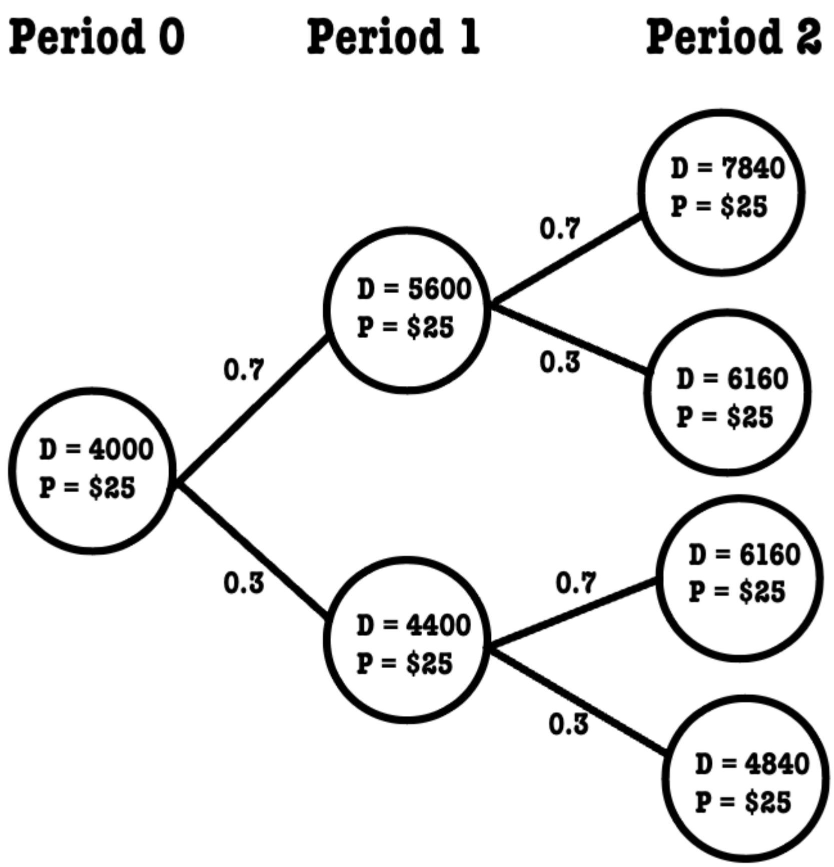
Question 1

a.



b.

First, we need to compute the probabilities of each demand state, represented as D_n, the set of all possible demands at period n.

Let i be the number of large increases in price, and p be the probability of having a large increase. At period n, the probability of having k large increases in price is binomially distributed as:

$$P(i = k \,|\, n) = \binom{n}{k} \cdot p^k (1-p)^{n-k}$$

Reciprocally, this means at period n we have $n-k$ small increases, and they each occur with probability $1-p$

Knowing that the demand at $n = 0$ is \$4,000, a large increase represents a 40% increase with probability $p = .7$ and a small increase represents a 10% increase with probability $p = .3$, we get the following table of probabilities and price-levels for each period:

n = 0

price	w/ prob.
4000	1

n = 1

price	w/ prob.
5600	.7
4400	.3

n = 2

price	w/ prob.
7840	.49
6610	.42
4840	.09

We want to find the expected discounted profit for each. Let P_n be the profit for period n, and D_n be the set of all possible demands for period n. The formula for when we haven't bought the oven would be:

$$\begin{aligned} E[P_{\text{old oven}}] &= \sum_{n=0}^{n=2} \left(\frac{1}{1+k}\right)^n E[P_n] \\ &= \sum_{n=0}^{n=2} \left(\frac{1}{1+k}\right)^n 25E[D_n \wedge 6000] \quad \text{(by linearity of the expectation)} \\ &= \sum_{n=0}^{n=2} \left[\left(\frac{1}{1+k}\right)^n \cdot 25 \sum^{D_n} (d \wedge 6000) P(D_n = d)\right] \\ &= \$362231.879 \end{aligned}$$

, where $(x \wedge y) = \min\{x, y\}$. We do this since we have no capacity to produce over 6000 units, and therefore our profit is capped at 25*6000.

Now, when the buy the oven at price O, we get the following formula:

$$\begin{aligned}
E[P_{\text{new oven}}] &= O + \sum_{n=0}^{n=2} \left(\frac{1}{1+k}\right)^n E[P_n] \\
&= O + \sum_{n=0}^{n=2} \left(\frac{1}{1+k}\right)^n 25E[D_n \wedge 10000] \quad \text{(by linearity of the expectation)} \\
&= O + \sum_{n=0}^{n=2} \left[\left(\frac{1}{1+k}\right)^n \cdot 25 \sum^{D_n} (d \wedge 10000) P(D_n = d)\right] \\
&= O + \$384624.630
\end{aligned}$$

.

So, to find the breakeven-price for the oven, we set the two expectations equal to each other. We get:

$$\begin{aligned}
E[P_{\text{old improved oven}}] &= E[P_{\text{new oven}}] \\
\$362231.879 &= O + \$384624.630 \\
O &= -22392.751
\end{aligned}$$

Therefore the maximum price the owner would want to pay for the new oven is \$22,392.751.

c.

Resolving, we get the following formula for the expected NPV of the old oven:

$$\begin{aligned}
E[P_{\text{old improved oven}}] &= \sum_{n=0}^{n=2} \left[\left(\frac{1}{1+k}\right)^n \cdot 25 \sum^{D_n} (d \wedge 6500) P(D_n = d)\right] \\
&= \$369448.040
\end{aligned}$$

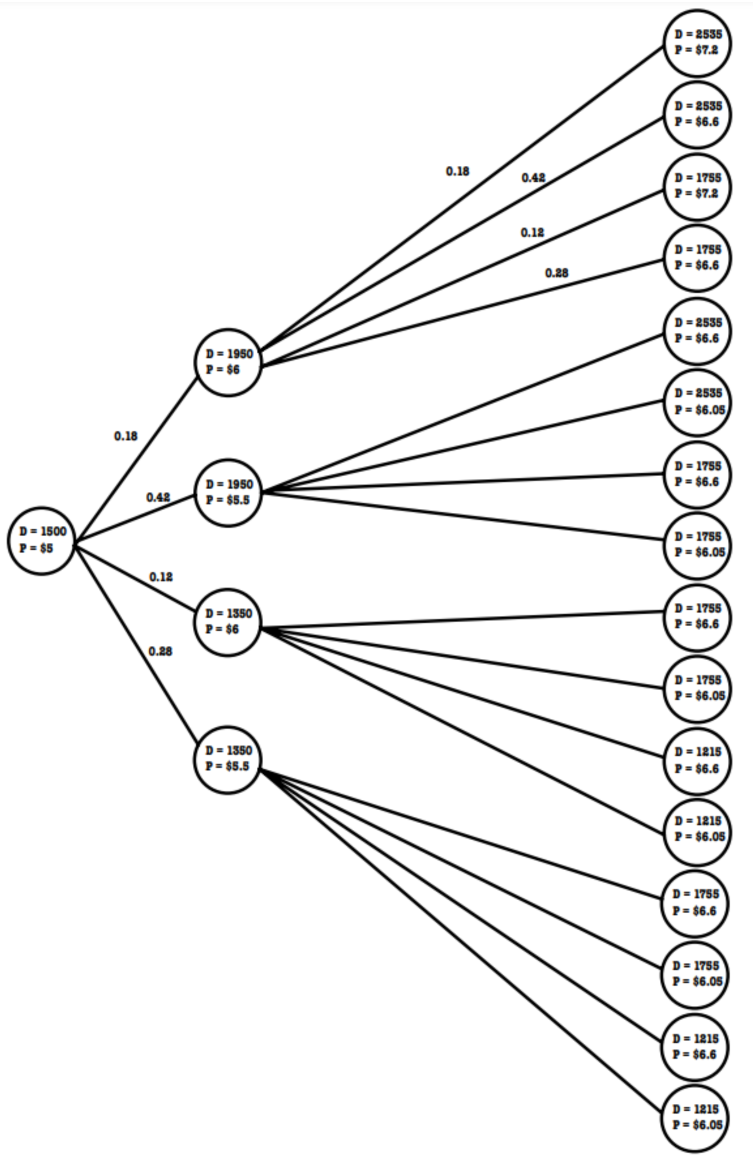
$E[P_{\text{new oven}}]$ doesn't change since neither the capacity of the oven nor its cost change, and it never was capacity-constrained. So we now have the following equation:

$$\begin{aligned}
E[P_{\text{old improved oven}}] &= E[P_{\text{new oven}}] \\
\$369448.040 &= O + \$384624.630 \\
O &= -15176.590
\end{aligned}$$

Therefore the maximum price the owner would want to pay for the new oven is \$15,176.590.

Question 2

a.


D = 1500
P = $5
0.18
0.42
0.12
0.28
D = 1950
P = $6
D = 1950
P = $5.5
D = 1350
P = $6
D = 1350
P = $5.5
0.18
0.42
0.12
0.28
D = 2535
P = $7.2
D = 2535
P = $6.6
D = 1755
P = $7.2
D = 1755
P = $6.6
D = 2535
P = $6.6
D = 2535
P = $6.05
D = 1755
P = $6.6
D = 1755
P = $6.05
D = 1755
P = $6.6
D = 1755
P = $6.05
D = 1215
P = $6.6
D = 1215
P = $6.05
D = 1755
P = $6.6
D = 1755
P = $6.05
D = 1215
P = $6.6
D = 1215
P = $6.05

b.

First, we need to compute the probabilities of each state, represented as (D_n, P_n), where D_n and P_n are the sets of all possible demands and prices at period n respectively.

Let i be the number of large increases in price, and p be the probability of having a large increase. At period n, the probability of having k large increases in price is binomially distributed as:

$$P(i = k \mid n) = \binom{n}{k} \cdot p^k (1-p)^{n-k}$$

Reciprocally, this means at period n we have $n-k$ small increases, and they each occur with probability $1-p$

Knowing that the price at $n = 0$ is \$5, a large increase represents a 20% increase with probability $p = .3$ and a small increase represents a 10% increase with probability $p = .7$, we get the following table of probabilities and price-levels for each period:

n = 0

price	w/ prob.
5	1

n = 1

price	w/ prob.
6	.3
5.5	.7

n = 2

price	w/ prob.
7.2	.09
6.6	.42
6.05	.49

We follow a similar logic for the demand. Let i be the number of increases in demand, and p be the probability an increase in demand. At period n, the probability of having k increases in demand is also binomially distributed.

Knowing that the demand at $n = 0$ is 1,500, an increase represents a 30% increase with probability $p = .6$ and a decrease represents a 10% decrease with probability $p = .4$, we get the following table of probabilities and demand-levels for each period:

n = 0

price	w/ prob.
1500	1

n = 1

price	w/ prob.
1950	.6
1350	.4

n = 2

price	w/ prob.
2535	.36
1755	.48
1215	.16

We want to find the expected discounted cost for each. Let C_n be the cost for period n. The formula for when we have no contract would be:

$$
\begin{aligned}
E[C] &= \sum_{n=0}^{n=2} \left(\frac{1}{1+k}\right)^n E[C_n] \\
&= \sum_{n=0}^{n=2} \left(\frac{1}{1+k}\right)^n E[D_n]E[P_n] \quad \text{(by independence)} \\
&= \sum_{n=0}^{n=2} \left[\left(\frac{1}{1+k}\right)^n \cdot \sum^{D_n} dP(D_n = d) \cdot \sum^{P_n} pP(P_n = p)\right] \quad = \$28611.5933
\end{aligned}
$$

For when we have a contract, we have a guaranteed cost of \$9,000 for the first 2,000 units. Therefore, if the demand is lower than 2,000, the variable cost is 0. So we get the following formula:

$$
\begin{aligned}
E[C] &= \sum_{n=0}^{n=2} \left(\frac{1}{1+k}\right)^n E[C_n] \\
&= \sum_{n=0}^{n=2} \left(\frac{1}{1+k}\right)^n \cdot \left[9000 + \sum^{D_n} (d-2000)^+ P(D_n = d) \cdot \sum^{P_n} pP(P_n = p)\right] \quad = \$27380.295
\end{aligned}
$$

,where $(x)^+ = \max\{x, 0\}$.
Therefore, we prefer having a contract.

c.

We now have the following formula for the new supplier:

$$
\begin{aligned}
E[C] &= \sum_{n=0}^{n=2} \left(\frac{1}{1+k}\right)^n E[C_n] \\
&= \sum_{n=0}^{n=2} \left(\frac{1}{1+k}\right)^n \cdot \left[7000 + \sum^{D_n} (d-1500)^+ P(D_n = d) \cdot \sum^{P_n} pP(P_n = p)\right] \quad = \$24854.267
\end{aligned}
$$

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So we prefer the new supplier over the other two options.